

CO4_AT2 – INDUSTRY PROBLEM SOLVING TASK

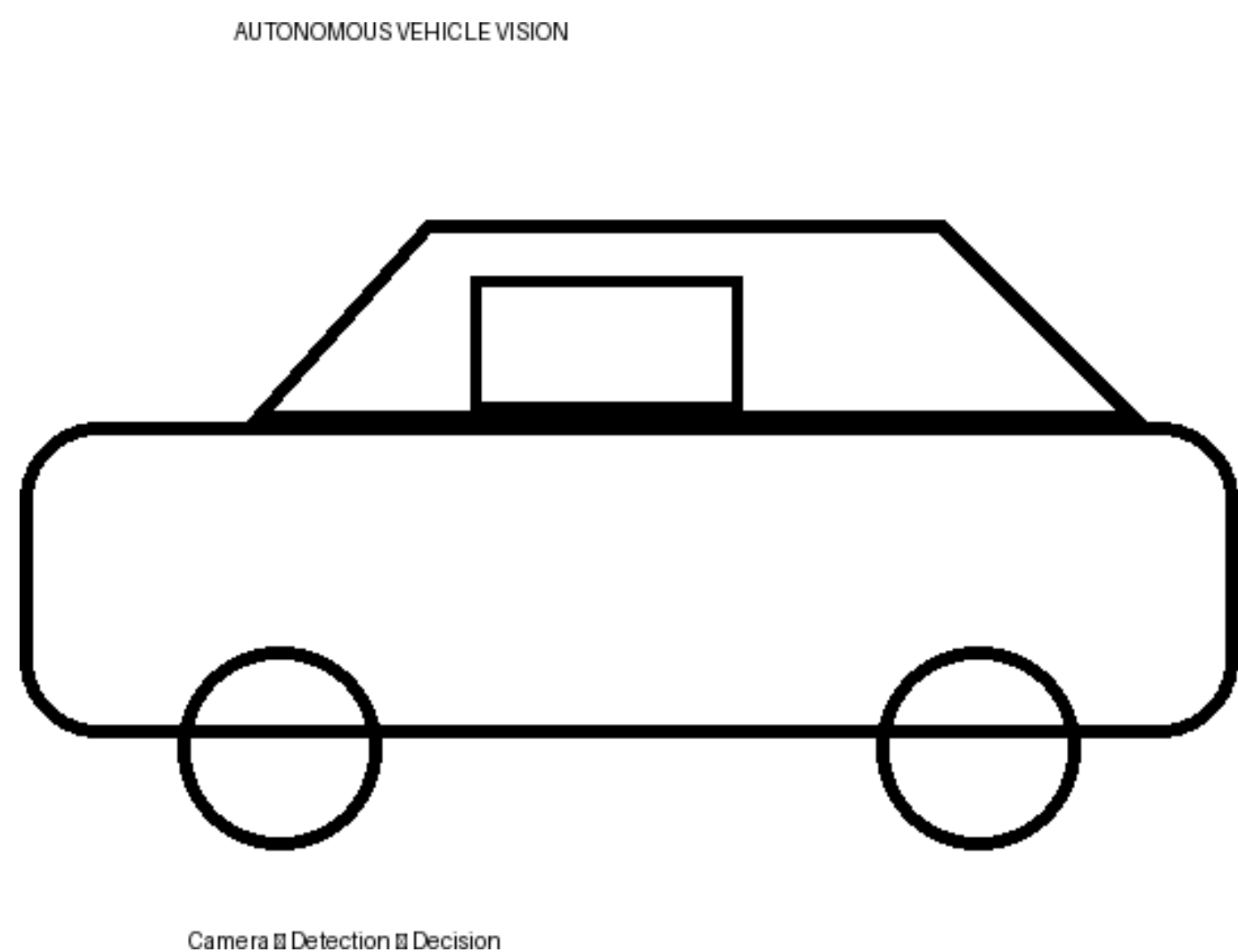
Expanded & Revised Industry Problem-Solving Solutions

This revised document keeps the original two industry cases but changes and expands the explanation. It includes additional reasoning, constraint analysis, practical deployment discussion, decision tables, and newly designed illustrations.

1. Autonomous Vehicle Vision System Selection

An automobile company is developing an autonomous vehicle detection system. The system must achieve at least 90% accuracy, maintain latency at or below 70 ms, and operate using moderate computational resources. Because these are mandatory constraints, a method that fails even one requirement cannot be selected.

New Illustration – Autonomous Vehicle Vision



Given Conditions

Constraint	Required
Accuracy	$\geq 90\%$
Latency	$\leq 70\text{ ms}$
Computation	Moderate

Available Methods

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High

Detailed Evaluation

Viola-Jones: Its 30 ms latency and low computation are advantageous for fast systems. However, 82% accuracy is below the mandatory 90% threshold. Since accuracy is a compulsory condition, the method is rejected.

YOLO: Its 93% accuracy exceeds the target, while 60 ms latency remains within the 70 ms limit. Moderate computation also matches the available resources. Therefore, all three constraints are satisfied.

Deep Learning: Its 97% accuracy is excellent, but 90 ms latency exceeds the allowed limit. In addition, high computational demand conflicts with the moderate-resource condition. It is therefore rejected

Advantages

- Very fast processing
- Low computational requirement
- Suitable for systems with limited hardware
- Simple implementation

Limitations

- Accuracy is only **82%**
- Does not meet the required **90% accuracy**
- Lower reliability for safety-critical detection

Decision

Although Viola-Jones satisfies the latency and computation requirements, it fails the

mandatory accuracy requirement.

Constraint Based Decision

Method	Accuracy	Latency	Computation	Decision
Viola-Jones	FAIL - 82%	PASS - 30 ms	PASS - Low	Reject
YOLO	PASS - 93%	PASS - 60 ms	PASS - Moderate	Select
Deep Learning	PASS - 97%	FAIL - 90 ms	FAIL - High	Reject

Additional Industry Interpretation

Autonomous vehicles require rapid processing because detected objects can affect immediate driving decisions. A high-accuracy model that takes too long to respond may not be suitable for a time-critical application. On the other hand, a very fast model with inadequate accuracy can produce unreliable detections. The best engineering solution therefore lies within the feasible region created by all requirements.

Deployment Considerations

The vehicle has moderate computational resources, so selecting a high-computation model could increase hardware requirements and system cost. YOLO avoids this issue while still providing accuracy above the required threshold. Its 60 ms latency also leaves a 10 ms margin below the maximum allowed 70 ms.

Final Answer

YOLO is selected. It satisfies all three constraints simultaneously: 93% accuracy $\geq$ 90%, 60 ms latency $\leq$ 70 ms, and moderate computation.

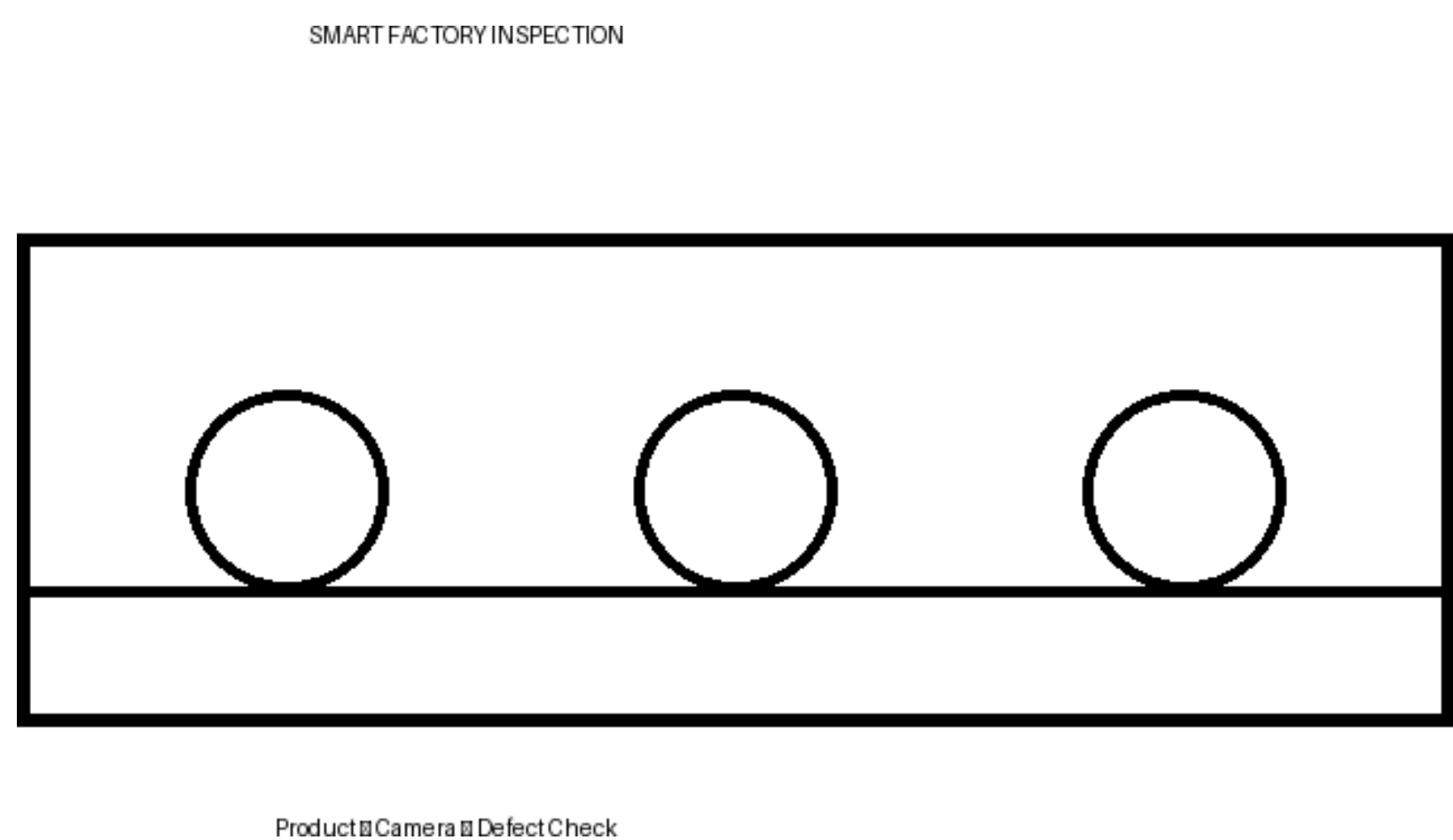
Conclusion

YOLO is the only method among the given alternatives that meets every mandatory requirement. This case demonstrates that industry decisions should consider the complete set of constraints rather than selecting a method solely because it has the highest accuracy.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to install a real-time computer vision system to identify defective products on a production line. The system must achieve at least 95% accuracy, process an image within 50 ms, and operate using moderate computational resources. The solution must therefore satisfy quality, speed and hardware constraints together.

New Illustration – Automated Inspection Station



Given Conditions

Constraint	Required
Accuracy	$\geq 95\%$
Processing Time	$\leq 50\text{ ms}$
Hardware	Moderate

Available Methods

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High

Detailed Evaluation

Traditional Image Analysis: Its 25 ms processing time and low computation make it fast and economical. However, 91% accuracy is below the required 95% , so it cannot provide the required inspection reliability.

YOLO: It reaches 96% accuracy, which is above the target. Its 45 ms processing time is below the 50 ms limit, and moderate computation matches the hardware. Therefore, it passes every condition.

Deep Learning: It provides the highest accuracy at 98%, but its 75 ms processing time exceeds the maximum 50 ms. Its high computational demand also violates the hardware condition

Important Factors

- Delayed detection can affect vehicle response.
- Low accuracy can result in missed objects.
- High computation can increase hardware requirements.
- A balanced algorithm is preferable when several constraints exist.

YOLO is selected because:

1. Accuracy is **93%**, above the required 90%.
2. Latency is **60 ms**, below the 70 ms limit.
3. Computational requirement is **moderate**.
4. It provides a practical balance between accuracy and speed.
5. It satisfies all mandatory constraints.

Autonomous Vehicle Comparison

Requirement	Viola-Jones	YOLO	Deep Learning
Accuracy $\geq$ 90%	✗	✓	✓
Latency $\leq$ 70 ms	✓	✓	✗

Moderate computation			
Overall	Reject	Select	Reject

Constraint-Based Decision and Practical Analysis

T	Method	Accuracy $\geq$ 95%	Time $\leq$ 50 ms	Moderate Computation	Decision
	Traditional Image Analysis	FAIL - 91%	PASS - 25 ms	PASS - Low	Reject
	YOLO	PASS - 96%	PASS - 45 ms	PASS - Moderate	Select

Production-Line Interpretation

A factory inspection system must operate continuously without becoming a bottleneck. If processing takes too long, products may accumulate while waiting for inspection. At the same time, insufficient accuracy can allow defective products to move to later stages or reach customers. Therefore, both detection reliability and processing speed are critical.

Why YOLO Gives the Best Balance

YOLO improves accuracy to 96% while keeping processing time at 45 ms. This provides a 5 ms margin under the 50 ms limit and satisfies the moderate-computation requirement. Deep Learning provides only a 2-percentage-point accuracy improvement over YOLO but violates both time and computation constraints.

Final Answer

YOLO is selected. It achieves 96% accuracy, 45 ms processing time and moderate computation, satisfying every mandatory requirement.

Overall Conclusion

Both industry cases demonstrate constraint-based problem solving. The best solution is not necessarily the model with the highest accuracy. In the autonomous vehicle case, YOLO satisfies accuracy, latency and computation requirements. In the smart factory case, YOLO satisfies accuracy, processing time and hardware requirements. Therefore, YOLO is the feasible and practical choice in both scenarios.